

- 

(A) 

(B)

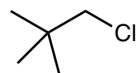
C1=CN=CNC1

(C) 

(D) 

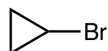
- (A) 

(B)



(C)

(D)



-

myo-inositol

The diagram shows a chemical structure of a nucleotide. It consists of a phosphate group (HOCH₂), a deoxyribose sugar, and a nitrogenous base. Four protons are labeled with letters A, B, C, and D, each enclosed in a circle. Proton A is attached to the phosphate group. Proton B is attached to the deoxyribose sugar. Proton C is attached to the nitrogenous base. Proton D is attached to the nitrogenous base.

(B) B

(C) C

(D) D

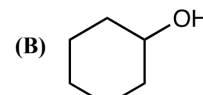
- (B)** 4

(C) 5

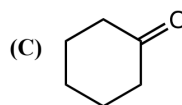
(D) 6

- (A) 

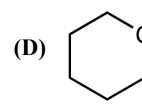
(A)



(B)



(C)



(D)

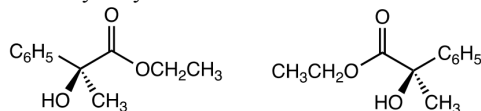
- *CC(=C)CC(C)C=CCC*

 CH_3 CH_3

1

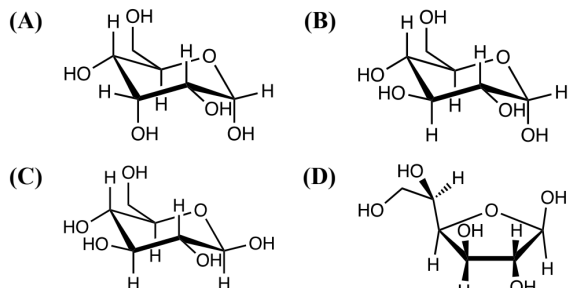
(D) 3-methyl-1,2-butadiene

11. In the presence of which catalysts will the two esters below hydrolyze at different rates from each other?



I. H_2SO_4 II. Pig liver esterase

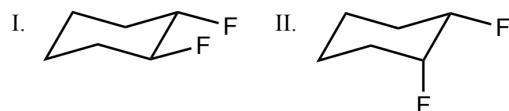
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
12. Which structure does not readily produce D-glucose when dissolved in acidic solution?



13. How many tertiary alcohols have the formula $\text{C}_5\text{H}_{12}\text{O}$?

(A) 1 (B) 2 (C) 3 (D) 4

14. Which difluorocyclohexanes are optically active?

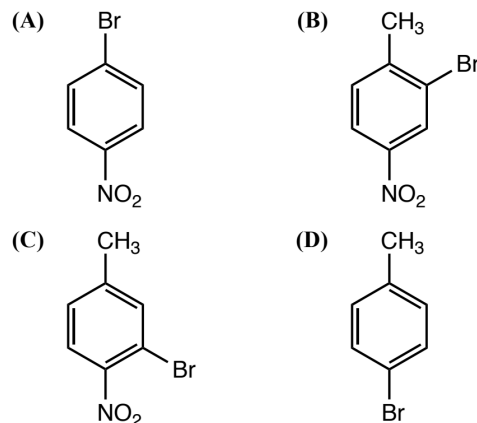
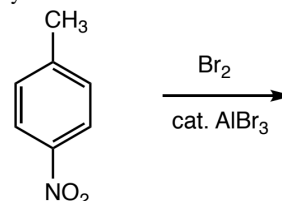


(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

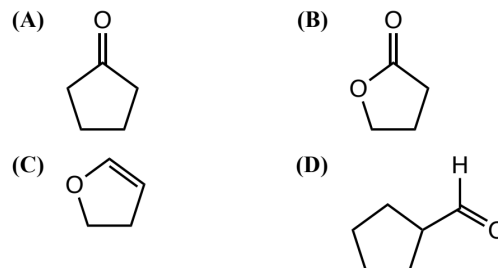
15. Which best describes the outcome of extracting a dilute solution of benzoic acid in diethyl ether with 5% aqueous NaHCO_3 ?

(A) Benzoic acid is present in the bottom layer.
(B) Benzoate ion is present in the bottom layer.
(C) Benzoic acid is present in the top layer.
(D) Benzoate ion is present in the top layer.

16. Which product would be obtained in the greatest quantity from bromination of 4-nitrotoluene using an aluminum bromide catalyst?



17. Which compound is least reactive toward ethylmagnesium bromide, $\text{CH}_3\text{CH}_2\text{MgBr}$?

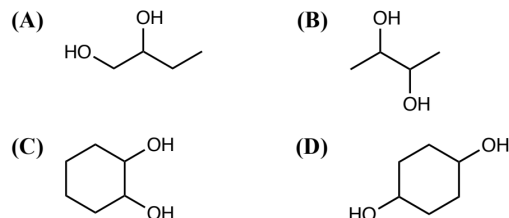


18. Which statements describe significant differences between RNA and DNA?

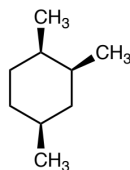
- I. The bases used in RNA include three purines and one pyrimidine, while those in DNA include two purines and two pyrimidines.
II. Hydrolysis of RNA frequently involves cyclic phosphate intermediates, while hydrolysis of DNA never does.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

19. Which molecule is achiral in all of its stereoisomeric forms?



20. How many methyl groups are axial in the most stable chair conformation of the trimethylcyclohexane shown below?

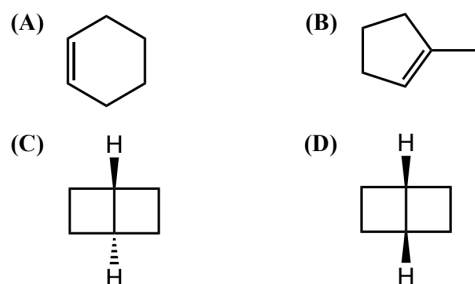


- (A) Zero (B) One (C) Two (D) Three

21. Which series lists the carboxylic acid derivatives in increasing order of reactivity in hydrolysis reactions?

- (A) $\text{CH}_3\text{CH}_2\text{CONH}_2 < \text{CH}_3\text{CH}_2\text{COOCH}_3 < \text{CH}_3\text{CH}_2\text{COCl}$
- (B) $\text{CH}_3\text{CH}_2\text{COOCH}_3 < \text{CH}_3\text{CH}_2\text{CONH}_2 < \text{CH}_3\text{CH}_2\text{COCl}$
- (C) $\text{CH}_3\text{CH}_2\text{CONH}_2 < \text{CH}_3\text{CH}_2\text{COCl} < \text{CH}_3\text{CH}_2\text{COOCH}_3$
- (D) $\text{CH}_3\text{CH}_2\text{COOCH}_3 < \text{CH}_3\text{CH}_2\text{COCl} < \text{CH}_3\text{CH}_2\text{CONH}_2$

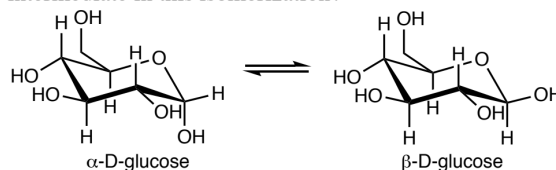
22. Which C_6H_{10} isomer has the most positive standard enthalpy of formation?



23. Which reagent reacts readily with butanal ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$), but not with 2-pentanone ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_3$)?

- (A) $\text{K}_2\text{Cr}_2\text{O}_7$ in aqueous H_2SO_4
- (B) I_2 in aqueous NaOH
- (C) CH_3MgBr in Et_2O
- (D) NaBH_4 in CH_3OH

24. α -D-glucose and β -D-glucose interconvert in aqueous solution at pH 7 (mutarotation). Which best describes the intermediate in this isomerization?



- (A) A planar oxacyclohexane
- (B) An acyclic aldohexose
- (C) A disaccharide
- (D) An achiral glycoside

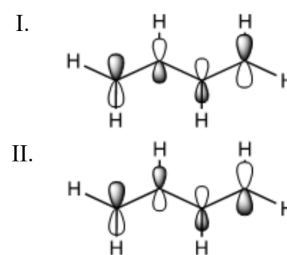
25. How many isomers have the formula $\text{C}_3\text{H}_6\text{Cl}_2$?

- (A) 4 (B) 5 (C) 6 (D) 7

26. Which compound produces a nitrile when treated with sodium cyanide in DMSO?

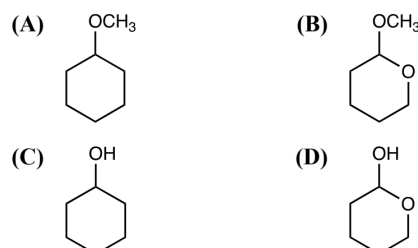
- (A) 2-Methylbutane
- (B) 3-Methyl-1-butanol
- (C) 3-Methyl-1-butene
- (D) 1-Bromo-3-methylbutane

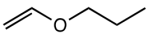
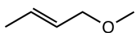
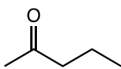
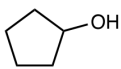
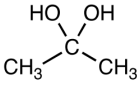
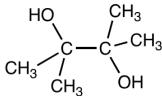
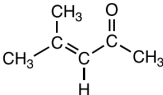
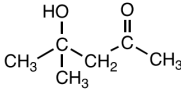
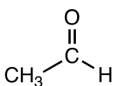
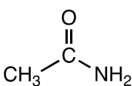
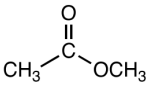
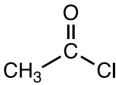
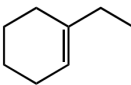
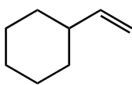
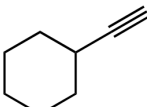
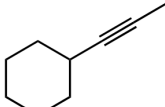
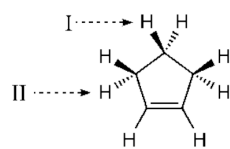
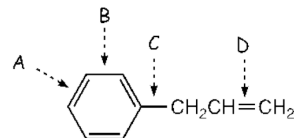
27. Which molecular orbitals are occupied in the ground state of 1,3-butadiene?



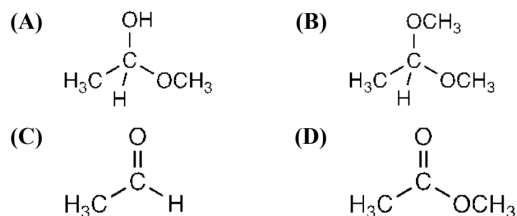
- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

28. Which compound reacts readily with water under acidic but not basic conditions?



29. Which $C_5H_{10}O$ isomer exhibits a strong infrared absorption band at 1717 cm^{-1} ?
- (A)  (B) 
 (C)  (D) 
30. The peptide hormone insulin becomes biologically inactive after being treated with a mild reducing agent. Which is the best explanation for this observation?
- (A) The reducing agent cleaves disulfide bonds, which alters the structure of the hormone.
 (B) The reducing agent cleaves peptide (amide) bonds, breaking the hormone into smaller polypeptides.
 (C) The reducing agent modifies the alcohol-containing side chains of some of the amino acids, decreasing the affinity of the hormone for its cellular receptor.
 (D) The reducing agent decreases the overall charge on the hormone, causing it to bind unselectively to cell membranes.
31. Which of these could have the formula $C_6H_{12}O$?
- I. An acyclic ester
 II. A cyclic ether
- (A) I only (B) II only
 (C) Either I or II (D) Neither I nor II
32. Which is the best explanation for the higher reactivity of conjugated dienes relative to non-conjugated alkenes in electrophilic addition reactions?
- (A) Conjugated dienes can form allylic cations on reaction with electrophiles while non-conjugated alkenes cannot.
 (B) Conjugated dienes have more potentially reactive sites than do non-conjugated alkenes.
 (C) The π bonding in conjugated dienes is weaker than the π bonding in non-conjugated alkenes.
 (D) Conjugated dienes are nonplanar while non-conjugated alkenes are planar.
33. Which compound will NOT form in the reaction of acetone (propanone) with aqueous base?
- (A)  (B) 
 (C)  (D) 
34. Which compound would have the lowest $C=O$ stretching frequency in its infrared spectrum?
- (A)  (B) 
 (C)  (D) 
35. Which compound reacts fastest with aqueous acid?
- (A)  (B) 
 (C)  (D) 
36. How many possible dipeptides can be formed using the twenty commonly occurring amino acids?
- (A) 40 (B) 200 (C) 210 (D) 400
37. Replacing which hydrogens with chlorine would give a chiral molecule?
- 
- (A) I only (B) II only
 (C) Either I or II (D) Neither I nor II
38. Which carbon-carbon bond is the shortest?
- 
- (A) A (B) B (C) C (D) D
39. What is the best description of the geometry at carbon in the transition state of the following reaction?
- $$OH^- + CH_3Br \rightarrow CH_3OH + Br^-$$
- (A) Trigonal planar (B) Tetrahedral
 (C) Trigonal bipyramidal (D) Square pyramidal

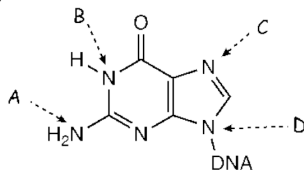
40. Which compound will NOT react readily in aqueous sodium hydroxide solution?



41. Treatment of acetyl chloride with an equimolar amount of diethylamine at room temperature in diethyl ether gives no more than a 50% yield of *N,N*-diethylacetamide. Which is the best explanation for this observation?

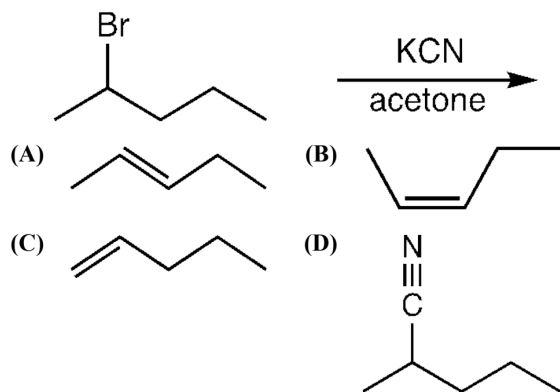
- (A) Half of the diethylamine is consumed by the neutralization of HCl produced in the reaction.
 (B) Hydrogen bonding between diethylamine molecules makes the reaction second-order in diethylamine.
 (C) Traces of water in the solvent convert at least half of the acetyl chloride into acetic acid.
 (D) Diethylamine is a strong base, but a weak nucleophile.

42. Guanine, one of the bases in DNA, is susceptible to attack by alkylating agents. Which nitrogen in guanine is most prone to alkylation?



- (A) A (B) B (C) C (D) D

43. What is the major organic product of the reaction of potassium cyanide with 2-bromopentane?



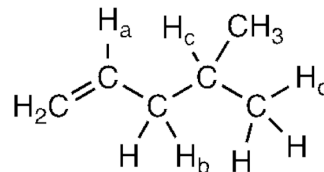
44. How many distinct compounds of the formula $C_5H_{11}Cl$ can be formed by free radical chlorination of 2-methylbutane?

- (A) 2 (B) 4 (C) 6 (D) 8

45. A compound with the formula C_5H_8O has exactly one π bond. How many rings does this compound have?

- (A) Zero
 (B) One
 (C) Two
 (D) It cannot be determined from the information given.

46. Which C–H bond has the smallest bond dissociation enthalpy?

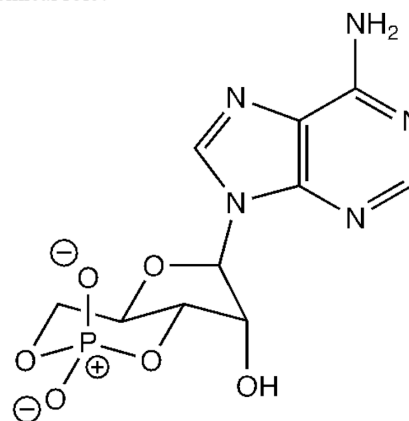


- (A) C–H_a (B) C–H_b (C) C–H_c (D) C–H_d

47. What compounds are formed on hydrolysis of a typical fat with aqueous sodium hydroxide (saponification)?

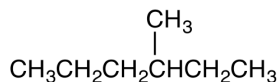
- (A) One equivalent each of an alcohol and the salt of a monocarboxylic acid
 (B) Three equivalents of alcohols and one equivalent of the salt of a tricarboxylic acid
 (C) One equivalent of a triol and three equivalents of the salt of monocarboxylic acids
 (D) One equivalent each of a triol and the salt of a tricarboxylic acid

48. The structure of cyclic adenosine monophosphate (cAMP) is shown below. What is its principal biochemical role?



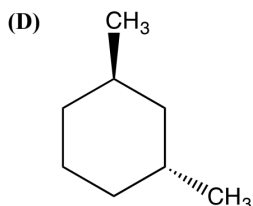
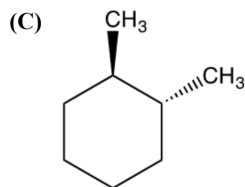
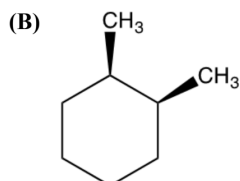
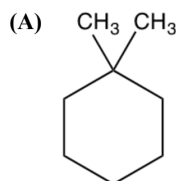
- (A) A signaling molecule
 (B) A source of energy
 (C) A synthetic precursor of DNA
 (D) A product of DNA degradation

49. Which statement about the branched hydrocarbon shown is correct?



- (A) Its IUPAC name is 4-methylhexane.
 (B) It is chiral.
 (C) It has six primary hydrogens.
 (D) Radical chlorination gives 1-chloro-4-methylhexane as the major product.

50. In which dimethylcyclohexane are both methyl groups equatorial in its most stable chair conformation?



51. How many distinct acyclic compounds have the formula C_5H_{10} ?

(A) 3 (B) 4 (C) 5 (D) 6

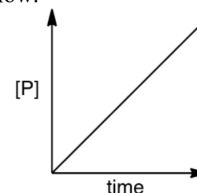
52. When acetaldehyde (CH_3CHO) reacts in methanol (CH_3OH) solution under basic conditions, which compound is NOT formed?

(A) $\text{CH}_3\text{CH}(\text{OH})(\text{OCH}_3)$ (B) $\text{CH}_3\text{CH}(\text{OCH}_3)_2$
 (C) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$ (D) $\text{CH}_3\text{CH}=\text{CHCHO}$

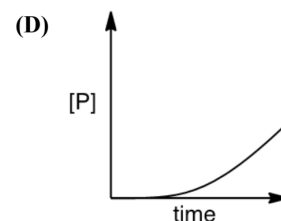
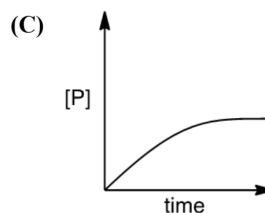
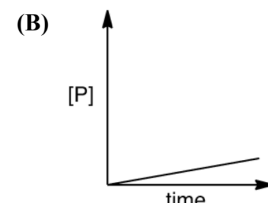
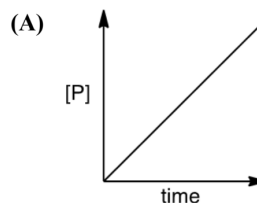
53. In what way does the reactivity of $\text{CH}_3\text{C}\equiv\text{CH}$ differ from $\text{CH}_3\text{CH}=\text{CH}_2$?

(A) Propyne is deprotonated by NaNH_2 , while propene is not.
 (B) Propyne does not react with bromine, while propene reacts readily with bromine.
 (C) Propyne undergoes catalytic hydrogenation over platinum, while propene does not.
 (D) Propyne is readily hydrated at $\text{pH} = 0$, while propene is not.

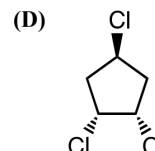
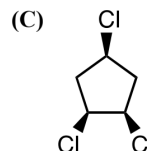
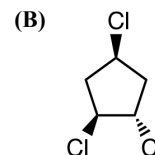
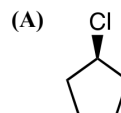
54. An enzyme catalyzes the transformation of a substrate into a product P, with the appearance of product over time as shown below:



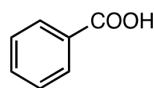
Under the same conditions, except in the presence of a competitive inhibitor of the enzyme, which graph best represents the appearance of product over time?



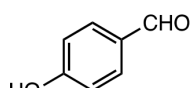
55. Which of the following molecules is chiral?



56. A chemist wishes to separate benzoic acid from 4-hydroxybenzaldehyde. Which is the best method to achieve this separation?

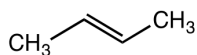


benzoic acid

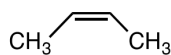


4-hydroxybenzaldehyde

- (A) Partitioning the mixture between diethyl ether and water
 (B) Partitioning the mixture between diethyl ether and 1 M aqueous NaHCO_3
 (C) Partitioning the mixture between diethyl ether and 1 M aqueous NaOH
 (D) Partitioning the mixture between diethyl ether and 1 M aqueous HCl
57. (*E*)-2-butene and (*Z*)-2-butene (shown below) each react with bromine to form compounds with the formula $\text{C}_4\text{H}_8\text{Br}_2$. What is the relationship between the products?

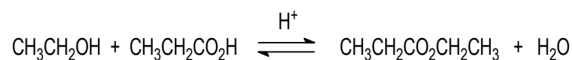


(*E*)-2-butene



(*Z*)-2-butene

- (A) Structural isomers (B) Enantiomers
 (C) Diastereomers (D) Identical
58. What is the role of the acid catalyst in the Fischer esterification reaction below?



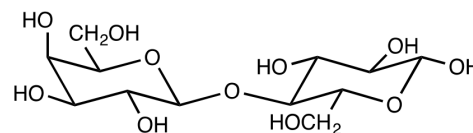
- (A) Shifts the equilibrium in the right-hand direction
 (B) Neutralizes the base formed as a side product in the reaction
 (C) Converts ethanol to a more reactive nucleophile
 (D) Converts propanoic acid to a more reactive electrophile

59. The isoelectric point of a protein is the pH at which it is electrically neutral. Which mutation of an amino acid $\text{NH}_2\text{CHRCOOH}$ in the protein would have the greatest effect on its isoelectric point, assuming that the mutation does not significantly affect the protein's overall structure?

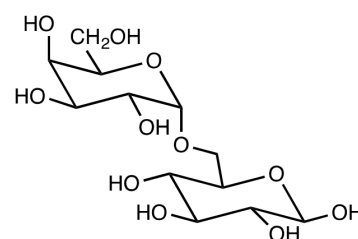
- (A) Serine ($\text{R} = \text{CH}_2\text{OH}$) $\rightarrow$ Lysine ($\text{R} = \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$)
 (B) Glutamine ($\text{R} = \text{CH}_2\text{CH}_2\text{CONH}_2$) $\rightarrow$ Methionine ($\text{R} = \text{CH}_2\text{CH}_2\text{SCH}_3$)
 (C) Isoleucine ($\text{R} = \text{CH}[\text{CH}_3]\text{CH}_2\text{CH}_3$) $\rightarrow$ Valine ($\text{R} = \text{CH}[\text{CH}_3]_2$)
 (D) Alanine ($\text{R} = \text{CH}_3$) $\rightarrow$ Glycine ($\text{R} = \text{H}$)

60. Hydrolysis of which disaccharide with dilute acid gives only a single type of monosaccharide as a product?

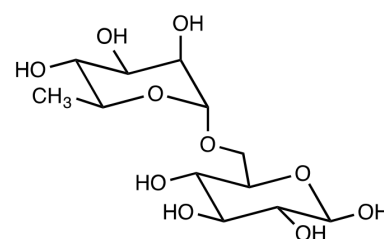
- (A) Lactose,



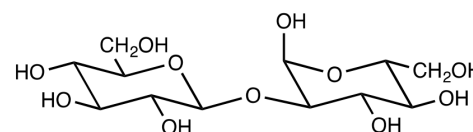
- (B) Melibiose,



- (C) Rutinose,



- (D) Sophorose,



61. A student wishes to prepare ethyl acetate from the reaction of ethanol and acetic acid. To be successful, this reaction requires

- (A) an acidic catalyst. (B) a basic catalyst.
 (C) an oxidizing agent. (D) a reducing agent.

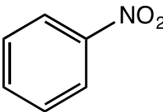
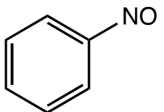
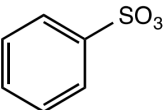
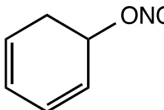
62. Which alkyl halide reacts most rapidly with aqueous sodium hydroxide solution?

- (A) CH_3Cl (B) CH_3I
(C) $(\text{CH}_3)_3\text{CCH}_2\text{Cl}$ (D) $(\text{CH}_3)_3\text{CCH}_2\text{I}$

63. How many isomers are there with the formula C_6H_{14} ?

- (A) 3 (B) 4 (C) 5 (D) 6

64. What is the product of the reaction of benzene with a solution of nitric acid in sulfuric acid at 50°C ?

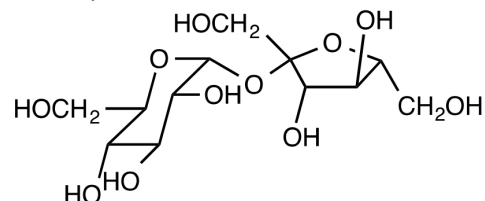
- (A)  (B) 
(C)  (D) 

65. An enzyme catalyzes the hydrolysis of an ester with a certain activity, but this activity is lost in a 3 M urea solution. What is the most likely explanation for the loss of activity?

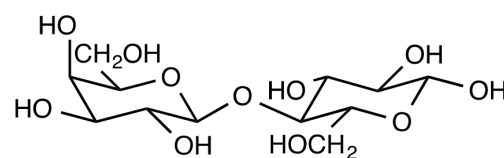
- (A) Urea binds to the active site of the enzyme competitively with the substrate.
(B) Urea causes the cleavage of the peptide bonds in the enzyme.
(C) Urea causes the enzyme to denature and lose its specific three-dimensional shape.
(D) Urea reacts with disulfide bonds in the enzyme.

66. Tollens' reagent, basic diamminesilver(I) solution, gives a positive test (in the form of a silver mirror) in the presence of aldehydes. Which sugars give a positive Tollens' test?

I. Sucrose,



II. Lactose,



- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

67. What is the formula for ethyl propanoate?

- (A) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$
(B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$
(C) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$
(D) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_2\text{OCH}_3$

68. Which substance can exist in enantiomeric forms?

- (A) 3-methylpentane (B) 2-methylpentane
(C) 3-methyl-1-butanol (D) 2-methyl-1-butanol

69. Which of the following is a difference between benzene and cyclooctatetraene?

- (A) Benzene reacts rapidly with bromine and cyclooctatetraene does not.
(B) Benzene contains sp^2 -hybridized carbon atoms and cyclooctatetraene does not.
(C) Benzene has the formula C_nH_n and cyclooctatetraene does not.
(D) Benzene is planar and cyclooctatetraene is not.

70. Which base is found in RNA, but not DNA?

- (A) Adenine (B) Guanine
(C) Thymine (D) Uracil

71. A triglyceride is formed from
- one molecule of glycerin and two molecules of fatty acid.
 - one molecule of glycerin and three molecules of fatty acid.
 - three molecules of glycerin and one molecule of fatty acid.
 - two molecules of glycerin and one molecule of fatty acid.

72. Which of these amino acids has the most hydrophobic side chain at neutral pH?

- Cysteine, $\text{HSCH}_2\text{CH}(\text{NH}_2)\text{COOH}$
- Glutamine, $\text{H}_2\text{NC}(\text{O})\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$
- Leucine, $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_2)\text{COOH}$
- Lysine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$

73. Mild oxidation of 1-propanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$, leads to the formation of

- propanal, $\text{CH}_3\text{CH}_2\text{CHO}$.
- propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.
- 2-propanone, CH_3COCH_3 .
- dipropyl ether, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{CH}_3$.

74. Which is the principal cation within cellular fluid?

- Na^+
- K^+
- Mg^{2+}
- Ca^{2+}

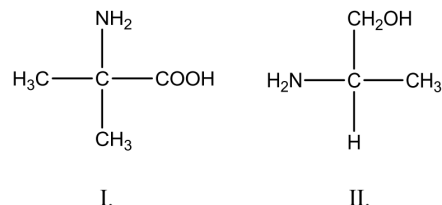
75. The shape of a chain segment of polypeptide chains in a protein is known as its

- primary structure.
- secondary structure.
- tertiary structure.
- quaternary structure.

76. Which property that polyacetylene exhibits is unusual for an organic polymer?

- electrical conductivity
- flexibility
- high boiling point
- strength

77. Which of the compounds below can exist as a pair of enantiomers?



- I only
- II only
- Both I and II
- Neither I nor II

78. Which type of dietary fat is the least healthy for humans?

- cis*-monounsaturated fat
- polyunsaturated fat
- saturated fat
- trans*-monounsaturated fat

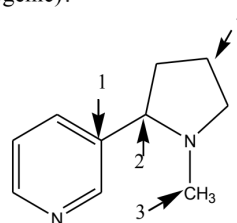
79. Which compound contains four co-linear carbon atoms?

- butane
- cis*-2-butene
- trans*-2 butene
- 2-butyne

80. Which substance can exist in boat and chair configurations?

- benzene
- cyclohexane
- cyclopentadiene
- naphthalene

81. Which of the numbered carbons in nicotine is chiral (stereogenic)?

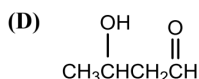
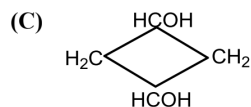
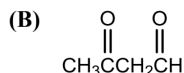
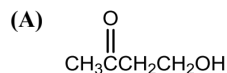


- 1
- 2
- 3
- 4

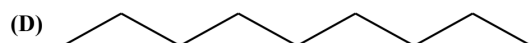
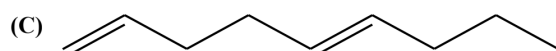
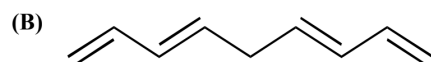
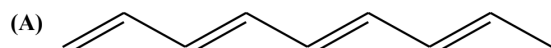
82. All of the following are monosaccharides except

- fructose
- glucose
- ribose
- sucrose

83. What is the product when two molecules of ethanal, CH_3CHO , undergo an aldol condensation?



84. Which molecule will have the longest wavelength absorbance maximum?



85. Which polymer is manufactured by condensation?

- (A) Polyethylene terephthalate
(B) Polypropylene
(C) Polystyrene
(D) Polyvinylchloride

86. How many non-cyclic compounds have the formula C_4H_8 ?

- (A) 1 (B) 2 (C) 3 (D) 4

87. Which is a Grignard reagent?

- (A) $\text{Ag}(\text{NH}_3)_2^+$ (B) $\text{C}_2\text{H}_5\text{MgBr}$
(C) $\text{FeBr}_3 + \text{Br}_2$ (D) LiAlH_4

88. A racemic mixture consists of equal quantities of

- (A) cis-trans isomers. (B) diastereomers.
(C) enantiomers. (D) structural isomers.

89. How many carbon-carbon double bonds are present in linolenic acid, $\alpha\text{-C}_{17}\text{H}_{29}\text{COOH}$?

- (A) 1 (B) 2 (C) 3 (D) 4

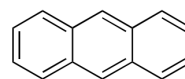
90. Recently scientists reported a bacterium that they believe incorporates arsenic into its DNA by substituting it for another element. Which element in DNA is arsenic most likely to replace?

- (A) carbon (B) nitrogen
(C) oxygen (D) phosphorus

91. Oxidation of a secondary alcohol with $\text{K}_2\text{Cr}_2\text{O}_7$ in sulfuric acid gives a product with which functional group?

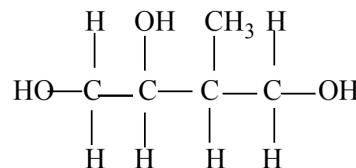
- (A) aldehyde (B) carboxylic acid
(C) ester (D) ketone

92. What is the formula for anthracene, which has the structure shown?



- (A) $\text{C}_{10}\text{H}_{10}$ (B) $\text{C}_{10}\text{H}_{20}$ (C) $\text{C}_{14}\text{H}_{10}$ (D) $\text{C}_{14}\text{H}_{14}$

93. How many chiral centers are present in the molecule shown?



- (A) 1 (B) 2 (C) 3 (D) 4

94. What is the major product when nitrobenzene is treated with a mixture of HNO_3 and H_2SO_4 ?

- (A) 1,1-dinitrobenzene (B) 1,2-dinitrobenzene
(C) 1,3-dinitrobenzene (D) 1,4-dinitrobenzene

95. Polysaccharides are biochemical molecules that consist of polymers of monosaccharide molecules (simple sugars). All of the following are classified as polysaccharides EXCEPT

- (A) cellulose. (B) fructose.
(C) glycogen. (D) starch.

96. Which metal is found in vitamin B_{12} ?

- (A) Co (B) Cu (C) Fe (D) Mn

97. Which compound exists in optically active forms?

- (A) $\text{CH}_3\text{CFCClCH}_3$ (B) $\text{CH}_2\text{FCH}_2\text{CH}_2\text{Cl}$
(C) $\text{CH}_2\text{FCHClCH}_3$ (D) $\text{CHF}_2\text{CH}_2\text{CH}_2\text{Cl}$

98. What product results when 2-butene reacts with chlorine?

- (A) 2-chlorobutane (B) 1,2-dichlorobutane
(C) 2,2-dichlorobutane (D) 2,3-dichlorobutane

99. Which chloroalkane undergoes substitution with OH^- exclusively by an $\text{S}_{\text{N}}1$ mechanism?

- (A) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ (B) $(\text{CH}_3)_3\text{CCl}$
(C) $\text{CH}_3\text{CH}_2\text{CHClCH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$

100. Which is a monosaccharide?

- (A) fructose (B) lactose
(C) maltose (D) sucrose

101. All of the following statements concerning benzene, C_6H_6 , are correct EXCEPT

- (A) Each carbon atom forms three sigma bonds.
(B) Each carbon is sp^2 hybridized.
(C) Pi electrons are delocalized over all 6 carbon atoms.
(D) Benzene forms cis and trans isomers when it reacts.

102. Which functional group is not commonly found in nucleic acids?

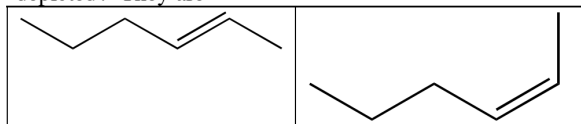
- (A) alcohol (B) amine
(C) carboxylic acid (D) dialkyl phosphate

103. Molecules of which type(s) do not contain a $\text{C}=\text{O}$ bond?

- I. amide
II. amine
III. ether
IV. ester

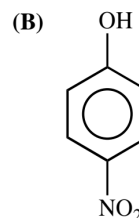
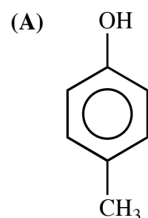
- (A) II only (B) I and IV only
(C) II and III only (D) III and IV only

104. What is the relationship between the two molecules depicted? They are



- (A) identical. (B) geometric isomers.
(C) structural isomers. (D) enantiomers.

105. For which compound is the $-\text{OH}$ group the most acidic?

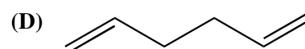
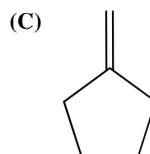
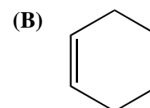
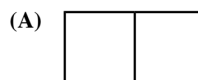


- (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (D) $(\text{CH}_3)_3\text{COH}$

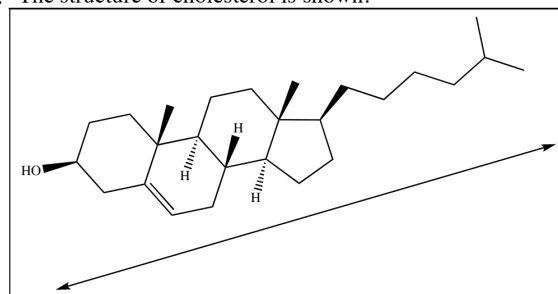
106. The most notable difference between a saturated and an unsaturated fat containing the same number of carbon atoms is that the saturated fat

- (A) melts at a higher temperature.
(B) melts at a lower temperature.
(C) releases much more energy when metabolized.
(D) releases much less energy when metabolized.

107. Which C_6H_{10} isomer is the least stable?



108. The structure of cholesterol is shown.



What is the approximate length of this molecule (as indicated by the scale bar)? ($1 \text{ \AA} = 1 \times 10^{-10} \text{ m}$)

- (A) $1.75 \text{ \AA}$ (B) $17.5 \text{ \AA}$
(C) $175 \text{ \AA}$ (D) $1750 \text{ \AA}$

109. Which molecule contains exactly eight carbon atoms?

- (A) benzoic acid (B) 2,3-dimethylhexane
(C) 3-ethylpentane (D) 3-methyloctane

110. Which formula represents an alkyne?
(Assume all are noncyclic.)
(A) C_2H_2 (B) C_2H_4 (C) C_5H_{10} (D) C_8H_{18}
111. How many compounds have the formula $C_2H_3Cl_3$?
(A) 2 (B) 3 (C) 4 (D) 5
112. Which is a condensation polymer?
(A) polyethylene (B) polyvinylchloride
(C) polystyrene (D) polyethylene terephthalate
113. What is the number of pi (π) bonds in trans-butenedioic acid ($C_4H_4O_4$)?
(A) 1 (B) 2 (C) 3 (D) 4
114. Cellulose and starch are biological polymers. Humans are able to digest starch but not cellulose. This difference is due **primarily** to a difference in the
(A) identity of the monomers in the two polymers.
(B) number of monomer units in the two polymers.
(C) orientation of the bonds joining the monomers.
(D) percentage of carbon in the two polymers.
115. How many unsaturated compounds have the formula C_4H_8 ?
(A) 3 (B) 4 (C) 5 (D) 6
116. Which compound is least soluble in water?
(A) $CH_3CH_2CH_2F$ (B) $CH_3CH_2CH_2NH_2$
(C) $CH_3CH_2CH_2OH$ (D) $CH_3CH_2CH_2COOH$
117. Which substance reacts most rapidly with water?
(A) C_6H_5Cl (B) $(CH_3)_3CCl$
(C) $(CH_3)_2CHCH_2Cl$ (D) $CH_3CH_2CH_2CH_2Cl$
118. What type of compound is formed by the mild oxidation of 2-pentanol?
(A) acid (B) aldehyde
(C) ester (D) ketone
119. Which species is lost during the formation of a disaccharide from a monosaccharide?
(A) CH_2 (B) CH_2O (C) CH_2OH (D) H_2O
120. How many structural isomers are possible for C_6H_{14} ?
(A) 2 (B) 3 (C) 4 (D) 5
121. Which is an ester?
(A) $CH_3COOCH_2CH_3$ (B) $(CH_3)_3COOC(CH_3)_3$
(C) CH_3OCH_3 (D) $(CH_3)_3CCOOH$
122. Which type of reaction is typical of aromatic compounds?
(A) addition
(B) free-radical substitution
(C) substitution by positively-charged reagents
(D) substitution by negatively-charged reagents
123. What is the IUPAC name of $(CH_3)_2CHCH=CHCH_3$?
(A) 1,2-methyl-isopropylethene
(B) 1,1-dimethyl-2-butene
(C) 1-isopropylpropene
(D) 4-methyl-2-pentene
124. Which compound can exist in optically active forms?
(A) $CH_3CH_2CH_2CH_2OH$ (B) $CH_3CH_2CH(OH)CH_3$
(C) $(CH_3)_2CHCH_2OH$ (D) $(CH_3)_3COH$
125. How many different tripeptides can be formed from the amino acids glycine, alanine and valine if each is used only once in each tripeptide?
(A) 3 (B) 4 (C) 5 (D) 6
126. All of the classes of compounds contain at least one oxygen atom EXCEPT
(A) esters (B) aldehydes
(C) ethers (D) alkynes
127. What is the most characteristic reaction of benzene?
(A) addition (B) polymerization
(C) reduction (D) substitution
128. Which organic acid is the strongest?
(A) $HCOOH$ (B) CH_3COOH
(C) $ClCH_2COOH$ (D) $ClCH_2CH_2COOH$
129. How many structurally isomeric alcohols have the formula C_4H_9OH ?
(A) one (B) two (C) three (D) four
130. Which compound can exist as two optical isomers?
(A) $ClHC=CHCl$ (B) meta- $C_6H_4Cl_2$
(C) CH_2ClBr (D) $CH_3CH(Cl)CH_2CH_3$

Organic/Biochemistry

131. Which type of dietary fat is currently considered the least harmful?

- (A) monounsaturated fat (B) polyunsaturated fat
(C) saturated fat (D) trans fat

132. How many different trichlorobenzenes, $C_6H_3Cl_3$, can be formed?

- (A) 1 (B) 2 (C) 3 (D) 4

133. What organic product is formed from the mild oxidation of a secondary alcohol?

- (A) acid (B) aldehyde
(C) ether (D) ketone

134. The compound with the formula, $H_2NCH_2CH_2COOH$, is best classified as a(n)

- (A) amide (B) amino acid
(C) fatty acid (D) nucleic acid

135. The reaction between which pair of reactants occurs the fastest for $[OH^-] = 0.010\text{ M}$?

- (A) $CH_3CH_2CH_2CH_2Cl + OH^-$
(B) $(CH_3)_3CCl + OH^-$
(C) $CH_3CH_2CH_2CH_2Br + OH^-$
(D) $(CH_3)_3CBr + OH^-$

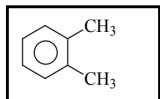
136. What is the major organic product formed from the reaction of $CH_3CH=CH_2$ and HCl ?

- (A) $CH_3CHClCH_3$ (B) $CH_3CH_2CH_2Cl$
(C) $CH_3CHClCH_2Cl$ (D) $CH_2ClCH=CH_2$

137. Fats and oils are formed from the combination of fatty acids with what other compound?

- (A) cholesterol (B) glucose
(C) glycerol (D) phenol

138. What is the formula for the compound?



- (A) C_8H_{10} (B) C_8H_{12}
(C) C_8H_{14} (D) C_8H_{16}

139. Which is most likely to react by an S_N1 mechanism?

- (A) CH_3Cl (B) $CH_3CHClCH_3$
(C) $(CH_3)_3CCl$ (D) C_6H_5Cl

140. Which substance will react most rapidly with $Br_2(aq)$?

- (A) benzene (B) chloropropane
(C) propanone (D) propene

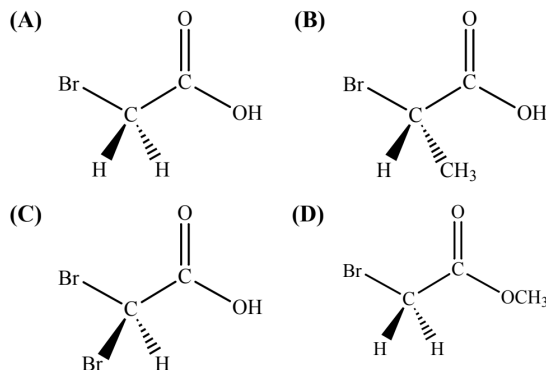
141. Which compound includes a carbon atom with an sp hybridized orbital?

- (A) benzene (B) butyne
(C) methyl chloride (D) phenol

142. Which compound has the highest vapor pressure at $25^\circ C$?

- (A) $CH_3CH_2CH_2CH_2OH$ (B) $CH_3CH_2CH_2OCH_3$
(C) $CH_3CH_2CH_2CH_2NH_2$ (D) $(CH_3)_3COH$

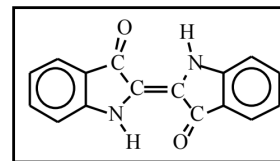
143. Which of the molecules can exist as optical isomers?



144. Which reaction is an oxidation? (Only the carbon-containing molecules are shown.)

- (A) $CH_2CH_2 \rightarrow CH_3CH_2OH$
(B) $CH_3CH_2OH \rightarrow CH_2CHO$
(C) $CH_3CH_2OH + HCOOH \rightarrow CH_3CH_2OOCH$
(D) $2\text{ }CH_3CH_2OH \rightarrow CH_3CH_2OCH_2CH_3$

Use this structure for the indigo molecule to answer questions 145 and 146.

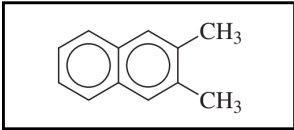
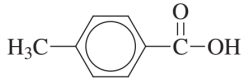
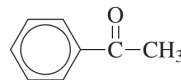
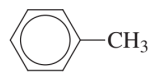
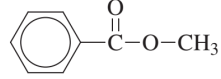


145. What is the molecular formula of indigo?

- (A) C_8HNO (B) $C_{16}H_{22}N_2O_2$
(C) $C_{16}H_{10}N_2O_2$ (D) $C_{16}H_{22}N_2O_2$

146. What is the hybridization of the carbon atoms bonded to oxygen?

- (A) sp (B) sp^2 (C) sp^3 (D) sp^3d

147. Aniline, $C_6H_5NH_2$, does not dissolve well in water. Which reagent could be used to increase its aqueous solubility?
- (A) 1 M HCl (B) 1 M NaOH
(C) diethyl ether (D) toluene
148. Which molecule reacts most rapidly with water?
- (A) $CH_3CH_2CH_2Cl$ (B) $CH_3CHClCH_3$
(C) $(CH_3)_2CHCH_2Cl$ (D) $(CH_3)_3CCl$
149. Which of these elements is found in hemoglobin?
- (A) Cr (B) Fe (C) Mg (D) Ni
150. Which compound is not correctly matched with its class name?
- (A) $HCOOH$, acid
(B) C_6H_5CHO , aldehyde
(C) $C_2H_5COCH_3$, ether
(D) $CH_3CHOHCH_3$, secondary alcohol
151. How many toluene derivatives have the formula C_7H_7Cl ?
- (A) 1 (B) 2 (C) 3 (D) 4
152. When the compounds CH_3COOH , C_2H_5OH and C_6H_5OH are arranged in order of increasing acidity in aqueous solution, which order is correct?
- (A) $C_2H_5OH < CH_3COOH < C_6H_5OH$
(B) $C_6H_5OH < CH_3COOH < C_2H_5OH$
(C) $CH_3COOH < C_6H_5OH < C_2H_5OH$
(D) $C_2H_5OH < C_6H_5OH < CH_3COOH$
153. Which can be used as a catalyst in an esterification reaction?
- I. NaOH II. H_2SO_4
- (A) I only (B) II only
(C) both I and II (D) neither I nor II
154. Which term best describes a carbocation?
- (A) electrophilic (B) free radical
(C) hydrophobic (D) nucleophilic
155. A racemic mixture has equal amounts of
- (A) alkanes and alkenes.
(B) *cis* and *trans* isomers.
(C) functional group isomers.
(D) enantiomers.
156. What is the molecular formula of this chemical structure?
- 
- (A) $C_{10}H_{12}$ (B) $C_{10}H_{14}$
(C) $C_{12}H_{12}$ (D) $C_{12}H_{14}$
157. Which is the formula for an alkyne?
- (A) C_2H_4 (B) C_3H_6
(C) C_3H_8 (D) C_4H_6
158. How many isomers have the formula C_3H_8O ?
- (A) 2 (B) 3 (C) 4 (D) 5
159. Which type of organic compound is most resistant to oxidation by acidified potassium dichromate?
- (A) acid (B) alcohol
(C) aldehyde (D) alkene
160. What product, in addition to water, is produced by this reaction?
- $$CH_3OH + C_6H_5COOH \rightarrow$$
- (A) 
- (B) 
- (C) 
- (D) 
161. How many carbon-carbon bonds are in a molecule of 2-methyl-2-butanol?
- (A) 2 (B) 3 (C) 4 (D) 5
162. Which molecule can exist as stereoisomers?
- (A) $CHF=CHF$ (B) $F_2C=CCl_2$
(C) CH_2F-CHF_2 (D) CF_3-CH_3

163. What are the most likely products in the reaction between $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and HI ?

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$ and H_2O
- (B) $\text{CH}_3\text{CH}_2\text{CH}_3$ and HOI
- (C) CH_3OH and $\text{CH}_3\text{CH}_2\text{I}$
- (D) $\text{ICH}_2\text{CH}_2\text{CH}_2\text{OH}$ and H_2

164. Addition polymers include

1. polyamide 2. polyethylene 3. polyester

- (A) 1 only (B) 2 only
- (C) 2 and 3 only (D) 1, 2 and 3

165. All of these are aromatic compounds *except*

- (A) hexene, C_6H_{12}
- (B) toluene, $\text{C}_6\text{H}_5\text{CH}_3$
- (C) *p*-dichlorobenzene, $\text{C}_6\text{H}_4\text{Cl}_2$
- (D) naphthalene, C_{10}H_8